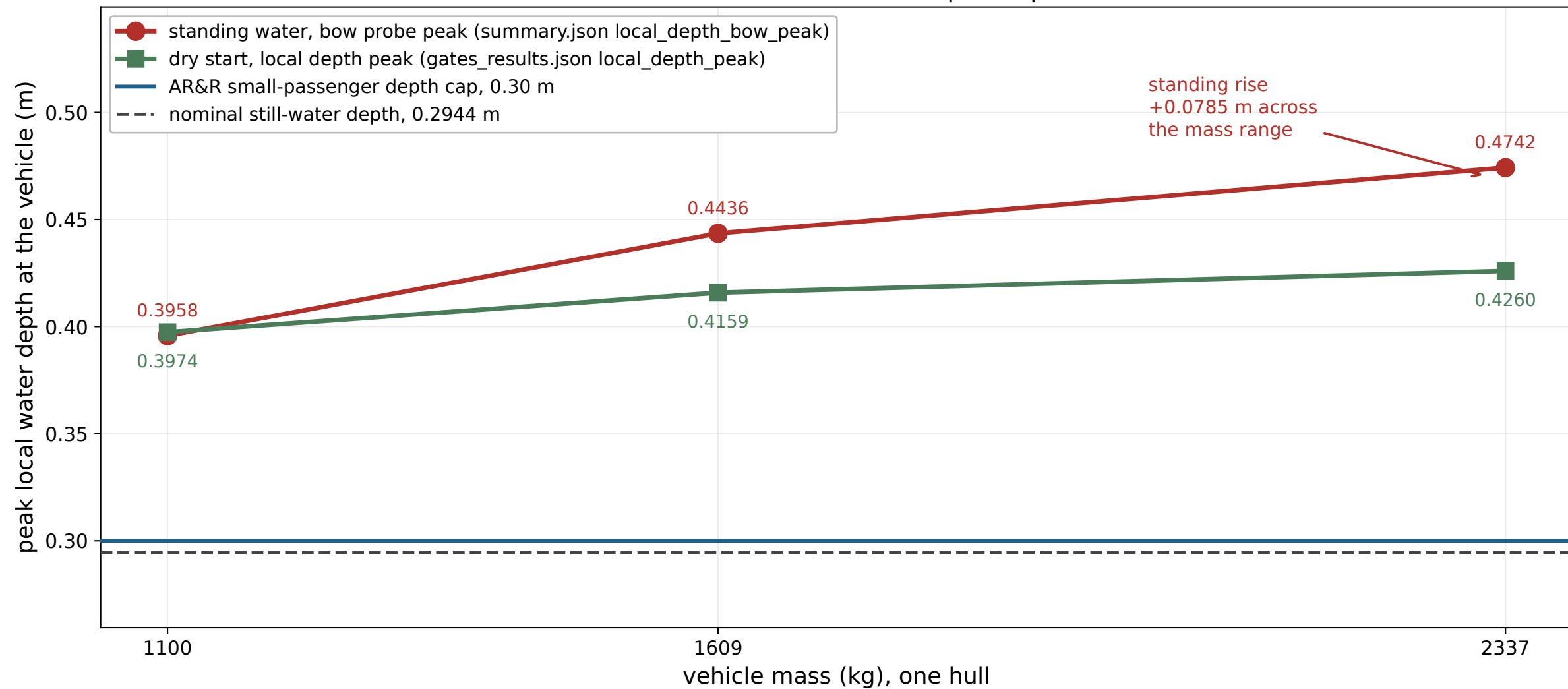


G4 The bow wave exceeds the AR&R depth cap in 19 of 20 runs



MEASUREMENT WINDOW. Standing water: the bow-probe series `local_depth_bow`, maximum over all 90 frames of the 3.0 s rollout, peaking at frames 50, 35, 24 for 1100, 1609 and 2337 kg respectively. Dry start: the local depth peak recorded in `gates_results.json`, peak frame 29. The two series are NOT the same probe, so the standing and dry curves are comparable in trend but not point-for-point. THE STORY: the nominal still-water depth is 0.2944 m, comfortably under the AR&R 0.30 m small-passenger cap, yet 19 of the 20 runs exceed that cap locally at the vehicle, and the exceedance grows with mass. The single exception is the slowest velocity-sweep run, $v = 0.5$ m/s, at 0.295361 m, which is not plotted here because this panel plots the mass series. Against each run's OWN class depth cap rather than the fixed 0.30 m small-passenger number, 16 of 20 exceed. All six runs plotted here do exceed. Standing water rises +0.0785 m from 0.3958 to 0.4742 m across the 2.1x mass range. A criterion evaluated on the undisturbed approach depth never sees this. TWO DRY-START PEAK MEASURES DISAGREE for the 1100 kg run: `gates_results.json local_depth_peak` gives 0.3974 m and `summary.json C2_local_depth_max` gives 0.4105 m. They are different probes over the same rollout. Only the `gates_results.json` series is plotted here; the C2 series is not mixed in. Source: `data/all_runs_inventory.csv` and `renders/yaris_render_s1/gates_results.json`.